Gayatri Vidya Parishad College of Engineering Autonomous

Artificial Intelligence and Machine Learning | Course Code: AI301

Machine Learning Foundations - Comprehensive Syllabus Lecture Notes

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Syllabus Focus: AI301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Machine Learning Foundations. Includes detail

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Mathematical Foundations & Linear Regression

" Syllabus Scope: Linear algebra, Maximum Likelihood Estimation MLE, Ordinary least squares, Ridge L2 and

Unit 2: Classification & Probabilistic Models

" Syllabus Scope: Logistic regression, Sigmoid derivation, Cross-entropy loss, Naive Bayes classifier, Ga

AI301: Machine Learning Foundations - Advanced Topics & Assessment

Academic Track: Artificial Intelligence and Machine Learning - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Support Vector Machines & Kernel Methods

" Maximum margin classifiers, Primal and Dual optimization, Karush-Kuhn-Tucker KKT conditions, Polynomial and

Unit 4: Decision Trees & Ensemble Learning

" Information gain, Gini impurity, ID3, C4.5, CART, Bootstrap Aggregation Bagging, Random Forests, AdaBoost, G

Unit 5: Unsupervised Learning & Dimensionality Reduction

" K-Means clustering, Hierarchical clustering, Expectation-Maximization EM, Principal Component Analysis PCA e

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Derive the closed-form normal equation theta = X^T X^-1 X^T y for multiple linear regression and prove
2. Formulate the dual optimization problem for Support Vector Machines with soft margins and explain slac
3. Formulate the mathematical derivation of Principal Component Analysis PCA using sample covariance matr

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Christopher M. Bishop - Pattern Recognition and Machine Learning Springer
* Kevin P. Murphy - Machine Learning: A Probabilistic Perspective MIT Press

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